

Who Can Turn Off Your Ethereum App?

A Full Stack Walk Away Test

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Ethereum File System



Building Cypherpunk Apps

Why make an app you can walk away from?

What happens when an app disappears?

Users lose their work and trust

Other developers can't rely on that functionality

Communities splinter

Software worth building on

For people

Keep years of useful work

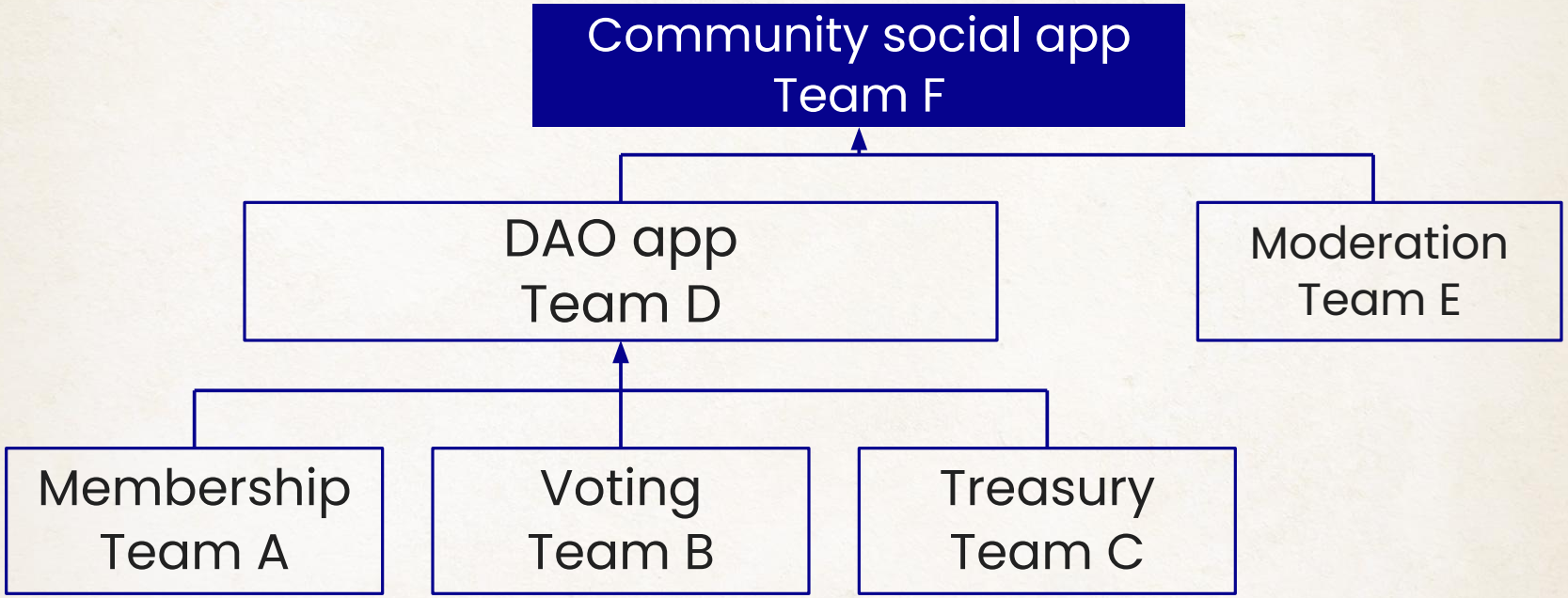
Choose better tools

For builders

Reuse dependable capabilities

Build bigger things together

Hyperstructures: build on each other



Independent teams work together via stable interfaces

A walk-away test

Single points of failure and control

The team disappears

Can users keep working?

Can others run the services?

Control becomes hostile

Can users keep their work?

Can they keep the rules they chose?

Choose one task your users rely on.

The application stack

Interface

- Names and discovery
- Runnable software
- Accounts and recovery

Data

- Storage and formats
- State and indexes
- Background work

Rules

- Contracts and governance
- Network and settlement



Interface

Finding and using the app

Find a saved release

Run it without your servers

Recover access without your help

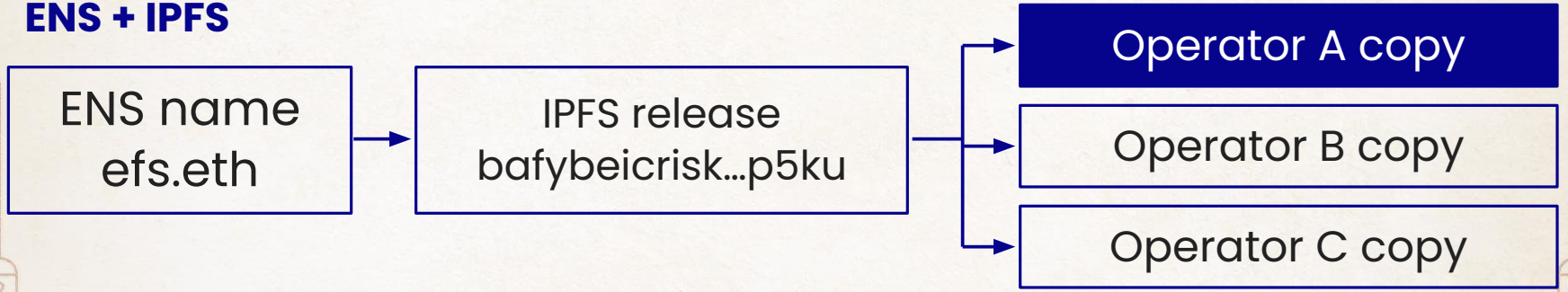
Can people find and use your app?

More than one way into the app

DNS + HTTP



ENS + IPFS



IPFS gateways: efs.eth.limo (or .link) or your own IPFS node

Name redirected? Open a saved, trusted release.

A complete runnable release

Keep every file the client needs to start

Incomplete release

index.html on IPFS

↓

Team server: app.js
Offline

Client cannot start

Complete release

IPFS release
index.html
app.js + styles + assets

↓

Client starts

Check: block the team's host, then reload the client

Access without the company

The login service closes. Can users still get in?



A route the user can control



Check access and recovery with the company offline



Data

Continuing the work

Is the data portable?

Keep the files and their meaning

Use them through another app

Let others run the supporting services

Data another app can understand

Example shared record

```
{
    id: note-17
    author: 0xDEADBEEF...
    text: Meeting at 10
    file: agenda.md
    vers: 2.0
}
```

Shared meaning

What each field means

Who can change it

How versions fit together

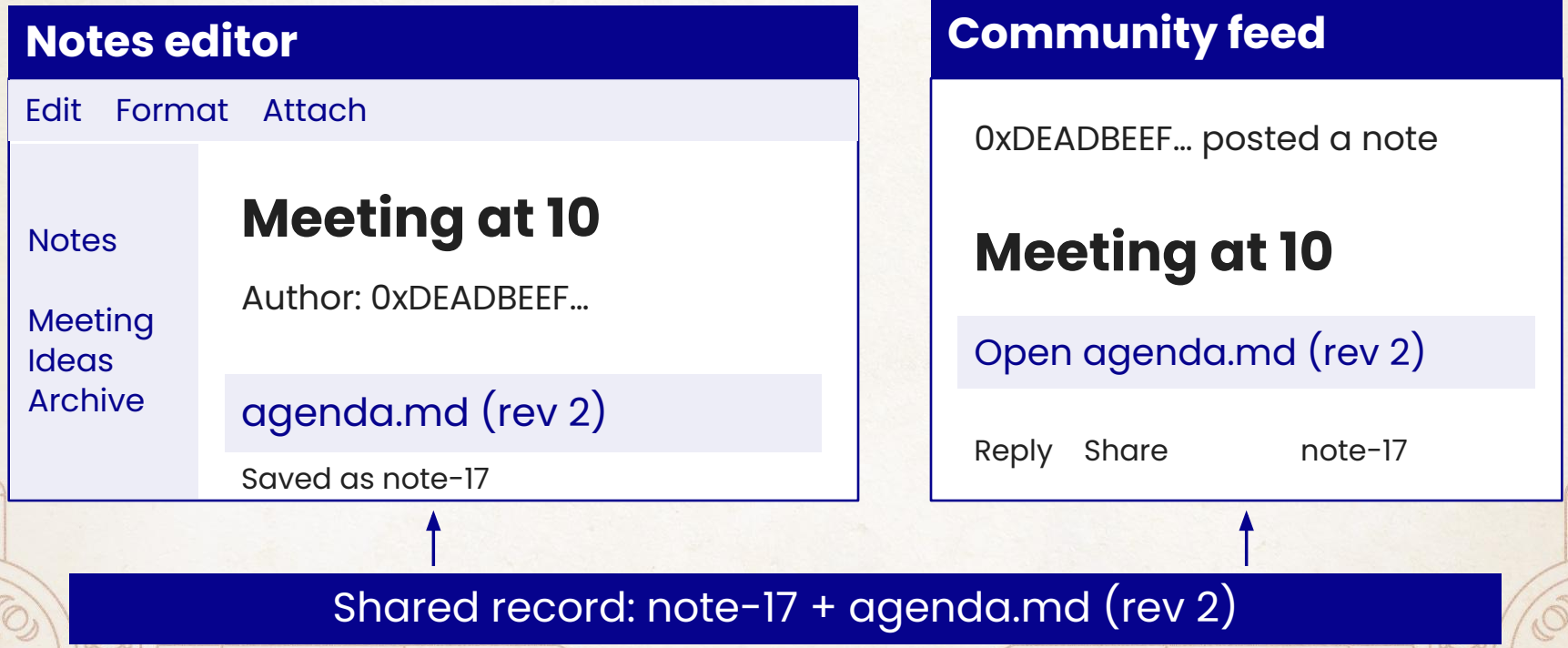
Who can read and decrypt it



Ethereum Attestation Service offers shared schemas

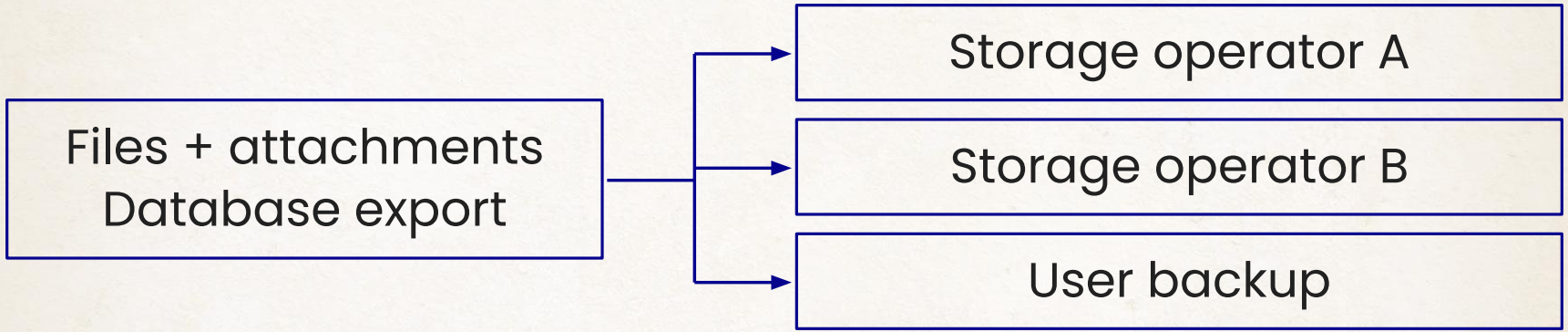
Your database can offer documented exports

Different tools. The same work.

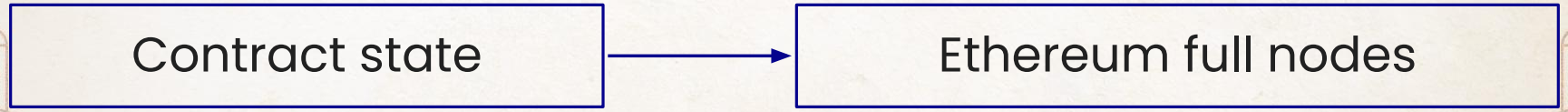


Who keeps a usable copy?

Files and databases

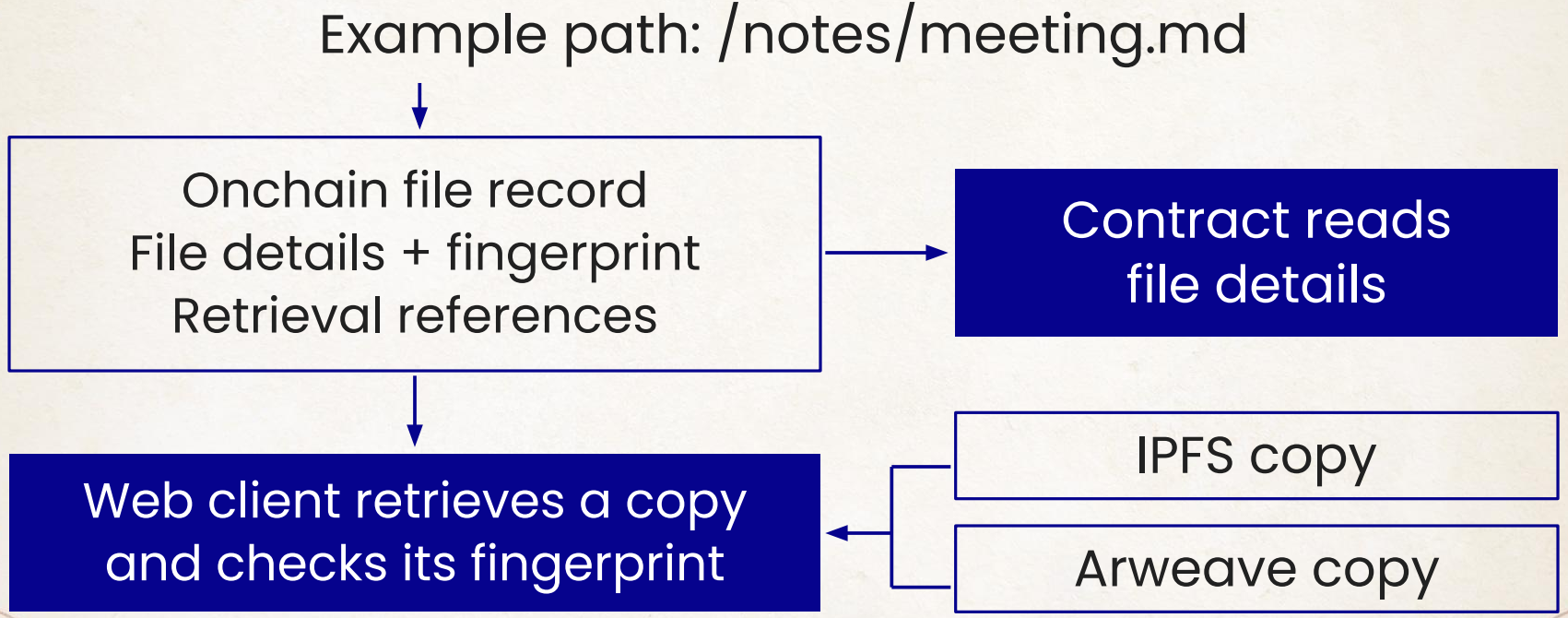


Onchain records



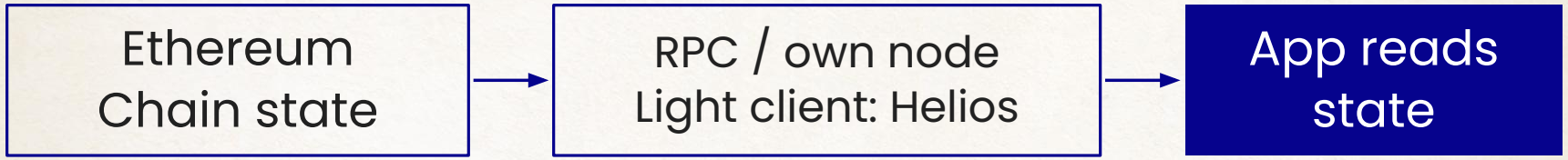
Check: restore useful work through an alternative

EFS: find a file and check a copy

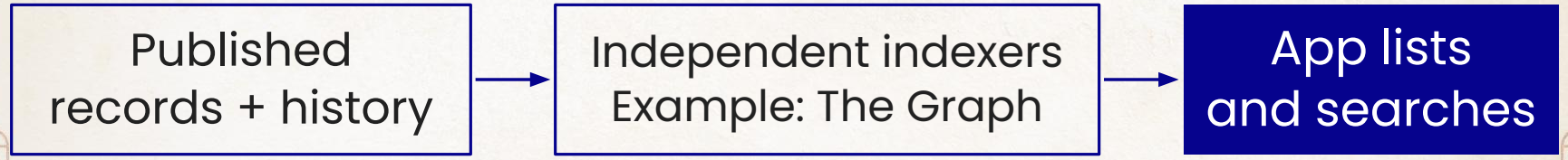


Reading and finding shared data

Read chain state



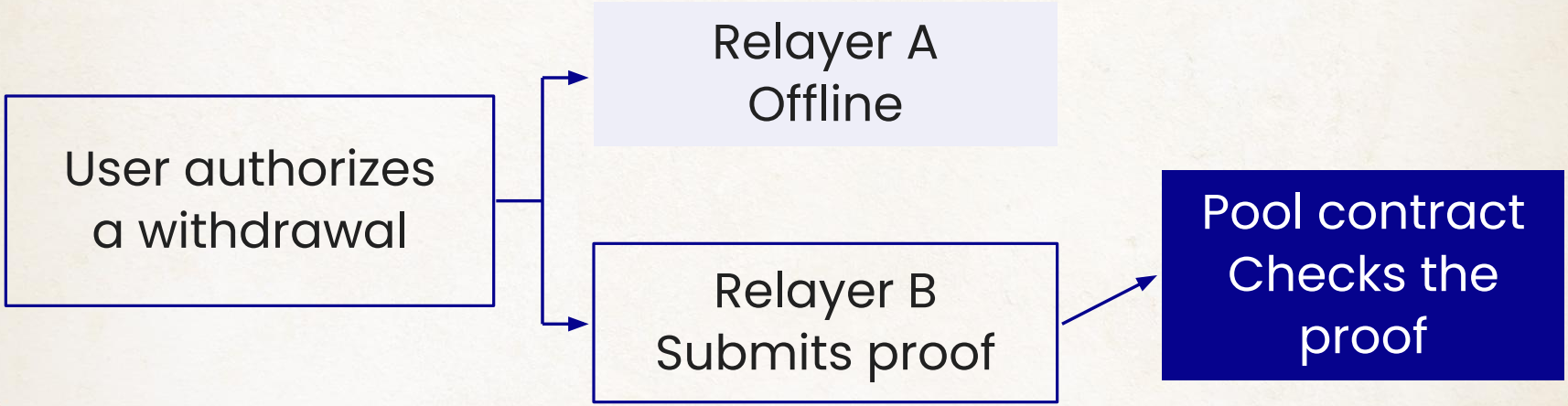
Find records in lists and searches



Check: change providers, rebuild a view and check its freshness

Who sends the next transaction?

Tornado Cash: replaceable relayers



Check: could another operator complete your background job?



Rules

Keeping dependable rules

Stable core contracts

User controlled app upgrades

Clear rules for shared governance

Who chooses the changes?

Immutable contracts

UNIX mindset: Do one thing and do it well

A stable core

Keep the contract code fixed

Let other apps build on it

Room to improve

Deploy a new version

Let users choose to migrate

Uniswap v2 core: fixed code, shared exchange logic

The network beneath the app

Ethereum and L1

Keep a working node route

Retain the history you need

Understand inclusion and finality

L2 and settlement

Know who orders and proves

Find the required chain data

Practice the recovery route

Someone else can take over

Original team offline

Public handoff

Software + license
Data + instructions
Configuration + costs

→ Independent operator

Their own keys and hosting

→ Replacement service

Users can switch to it

Check: can someone start it without contacting you?



Putting this into practice

A walk away test for your app

Your walk away checklist

A user can:

- ☑ Run a saved app release
- ☑ Recover access and keys
- ☑ Restore usable data
- ☑ Choose another client

Another operator can:

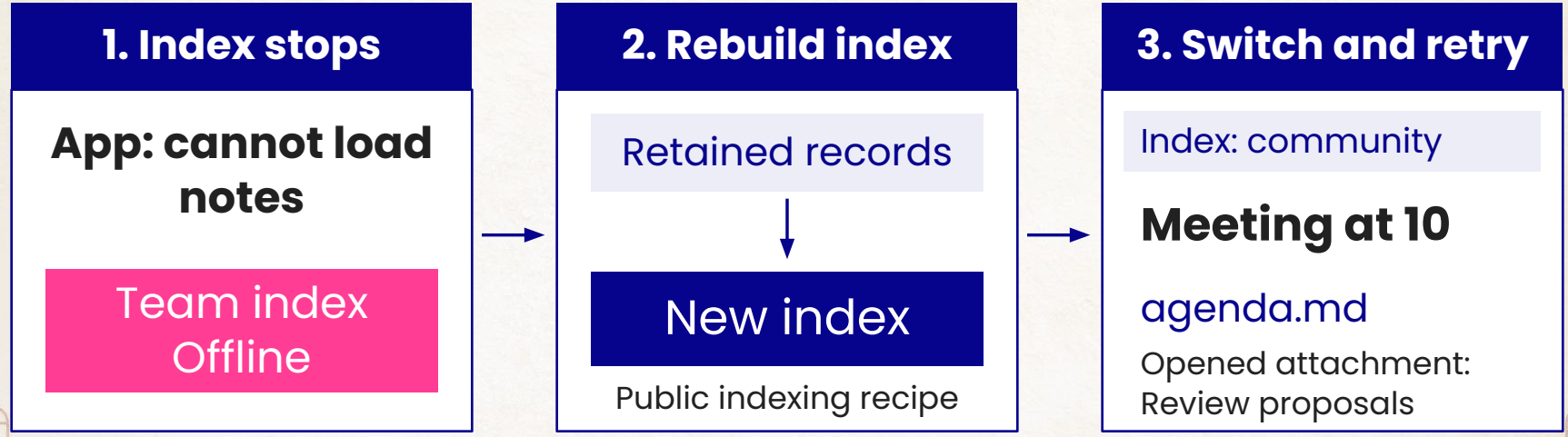
- ☑ Switch RPCs, rebuild views
- ☑ Run workers, keep copies
- ☑ Inspect admin powers
- ☑ Use network recovery

Who can upgrade, pause, redirect or change permissions?

Try it without your team. Improve one dependency.

A walk away test in practice

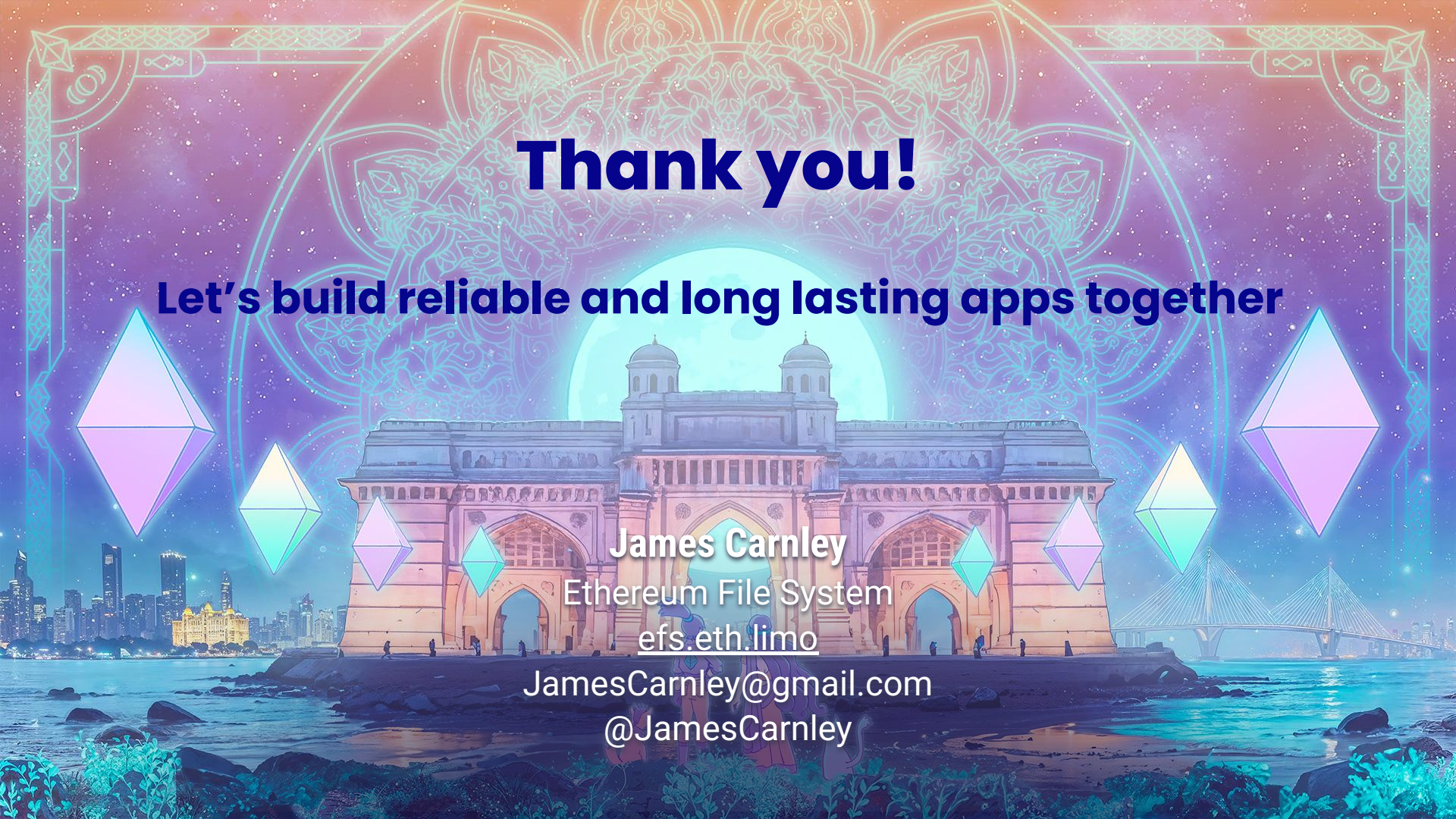
Example: find a note and open its attachment



Change the app's index endpoint to the replacement

Check: the same note appears and its attachment opens

Our work can outlive us


Thank you!
Let’s build reliable and long lasting apps together
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